AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM341
Computer Vision

Mid-term Examination MTA - Makeup
Fall Semester 2025/2026

Date: 6/Dec. /2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed: 120 Minutes**

Instructions:

- Total Marks: 30
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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Part – A (5 Marks)

1. The main advantage of the Sobel operator is:
   a) Detecting color in images
   b) Reducing image size
   c) Enhancing edges by considering neighboring pixels
   d) Rotating the image

2. In image processing, the term 'kernel' refers to:
   a) The main part of the image
   b) A small matrix used for convolution
   c) The RGB values
   d) A loss function

3. ----------contains random occurrences of black and white pixels
   a) Salt and pepper noise
   b) Impulse noise
   c) Gaussian noise
   d) All of the above

4. In computer vision, in the low level stage the output is an
   a) image
   b) features
   c) analysis
   d) None of the above

5. Closing is
   a) Dilation
   b) Erosion
   c) Dilation followed by Erosion
   d) Erosion followed by dilation

Part – B (10 Marks)

Question 1: (3 marks).

a) Explain the difference between edge detection and corner detection.
b) What is the purpose of using cv2.warpAffine() in OpenCV?

Question 2: (4 marks)

a) What is the purpose of applying an edge detection filter in image processing?
b) You are given a 5×5 grayscale image (matrix of pixel values) and asked to apply Sobel filter in the x-direction with the following kernel:

[[-1, 0, 1],
 [-2, 0, 2],
 [-1, 0, 1]]

Calculate the new value of the center pixel after applying the Sobel filter in the x-direction.

Assume the 3×3 region around the pixel is:

[[10, 10, 10],
 [10, 50, 10],
 [10, 10, 10]]

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Question 3: (3 marks)
The following code performs an affine transformation on an image. Fill in the blanks to complete the code that rotates the image 45 degrees around its center:

```
python
import numpy as np
import cv2 as cv

img = cv.imread('messi5.jpg', cv.IMREAD_GRAYSCALE)
assert img is not None, "file could not be read"

rows, cols = img.shape

Define rotation matrix
M = cv.getRotationMatrix2D(____, ____, _____)

Apply the rotation
dst = cv.warpAffine(img, M, (cols, rows))
```

a) Fill in the blanks to rotate the image 45 degrees around its center
b) Explain what cv.getRotationMatrix2D does.

Part – C (15 Marks)

Question 1: (5 marks)

a) What are Image Pyramids in image processing?
b) Explain the difference between Gaussian Pyramid and Laplacian Pyramid.
c) Write a simple Python code to generate both pyramids using OpenCV.

Question 2: (5 marks)

a) What is the purpose of converting the image to grayscale before applying the SIFT detector?
b) Explain what cv2.SIFT_create() and sift.detect() do in the context of feature detection.
c) Write a Python code snippet to display the image with keypoints after detecting them using SIFT.

Question 3: (5 marks)

a) Explain how the Prewitt operator is used for edge detection in image processing.
b) write a Python program that reads a grayscale image and applies the Prewitt filter in both the horizontal (x) and vertical (y) directions to detect edges. Then show the output.

```
Prewitt kernels are:
kernelx = np.array([[-1, 0, 1],
                    [-1, 0, 1],
                    [-1, 0, 1]])

kernely = np.array([[1, 1, 1],
                    [0, 0, 0],
                    [-1, -1, -1]])
```

End of questions

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